



ED32C-0547

Novel Student Teams Provide Authentic Geoscience Research Experience

E. Christa Farmer, J Bret Bennington, Antonios E. Marsellos, Jase E. Bernhardt, Gail K. Bennington
Hofstra University

Deborah K. Hecht
CUNY Graduate School



The Problem:

Due to a **projected shortage of geoscience professionals** now and in the near future (AGI, 2020), and the continuing **lack of diversity within the geosciences**, (AGI 2021), we need to increase the number and diversity of undergraduate students pursuing geoscience-related degrees, entering geoscience professions, and using geoscience skills to solve real-world problems as STEM professionals. These actions should increase the **innovation** within the geosciences (Hofstra et al., 2020).

Methods:

This National Science Foundation-funded project (award #1911514) creates **diverse student teams** and brings them together to pursue authentic geoscience research over an intensive 6-week summer program. Each "GEOTeam" (see **Figure 1**) includes a pre-service geoscience teacher, an undergraduate STEM major, a high school student, and a professor in the Hofstra University Geology, Environment, and Sustainability Department. Including high school students and pre-service teachers accomplishes several goals which broaden geoscience participation, including:

- Introducing students to geoscience research before they graduate from high school and become too invested in other career paths
- Introducing teachers to authentic geoscience research so they can share it with their students
- Providing opportunities for both undergraduate and pre-service teacher to mentor the high school student and draw them into the team and the geoscience research

Weekly workshops include safety training and basic research skills; and weekly seminars bring geoscience professionals from industry, non-profit organizations, and government to discuss geoscience careers in these various sectors. The program culminates in a poster session where the teams present their research. See **Figure 2** for an annual project timeline.

Project Goals Include:

1. Engaging GEOTeam members in **collaborative, authentic, scientific inquiry** doing geoscience research
2. Deepening interest in and knowledge of **geoscience thinking**
3. Introducing team members to **geoscience career possibilities**
4. Including **multiple members of underrepresented groups** each year
5. Strengthening the ability of pre-service teachers to **inspire their students** in the study of geoscience

Program Evaluation:

Deborah Hecht, of the City University of New York's Graduate School, provided external evaluation that included weekly surveys of all student participants in the summer programs in 2021 and 2022. One-on-one interviews of a few participating students were also conducted in 2021.

Table 1. Demographics of GEOTeams program participants by year.

	2021 Participants:	2022 Participants:
Women and Non-binary people:	7 out of 12	7 out of 12
People from Underrepresented Minority* (URM) groups:	5 out of 12	2 out of 12
People with a disability:	1 out of 12	0 out of 12

*according to the definition used by NCSES, 2019: blacks or African Americans, Hispanics or Latinos, and American Indians or Alaska Natives.



Acknowledgements:

We gratefully acknowledge all the GEOTeams participants. Most 2022 participants are pictured above, at the culminating October poster session.

We also gratefully acknowledge the National Science Foundation "Improving Undergraduate STEM Education: Pathways into Geoscience" program, which awarded us the funds to run this program (Award #1911514), and the leadership of Dr. Brandon Jones. The "hangouts" throughout the pandemic have been particularly helpful in the implementation of this project.

The support of the Hofstra University Sponsored Programs Office has been invaluable in administering this award, and we thank the wonderful folks there as well.

References:

- American Geosciences Institute, 2020. "Geoscience Workforce Projections 2019-2029." <https://www.americangeosciences.org/geoscience-currents/geoscience-workforce-projections-2019-2029>. Accessed 14 November 2022.
- American Geosciences Institute, 2021. "Status of Recent Geoscience Graduates 2021." <https://www.americangeosciences.org/static/files/GraduateSurvey2021.pdf>. Accessed 14 November 2022.
- Hecht, Deborah, 2022. Unpublished DUE # 1911514 Interim Evaluation Report Year 2021-2022.
- Hofstra, B., et al., 2020. "The Diversity-Innovation Paradox in Science." *PNAS* 117(17): 9284-9291.
- National Center for Science and Engineering Statistics, 2019. "Women, Minorities, and Persons with Disabilities in Science and Engineering." <https://nces.nsf.gov/pubs/nsf19304/digest/introduction> Accessed 21 November 2022.

People on GEOTeams:

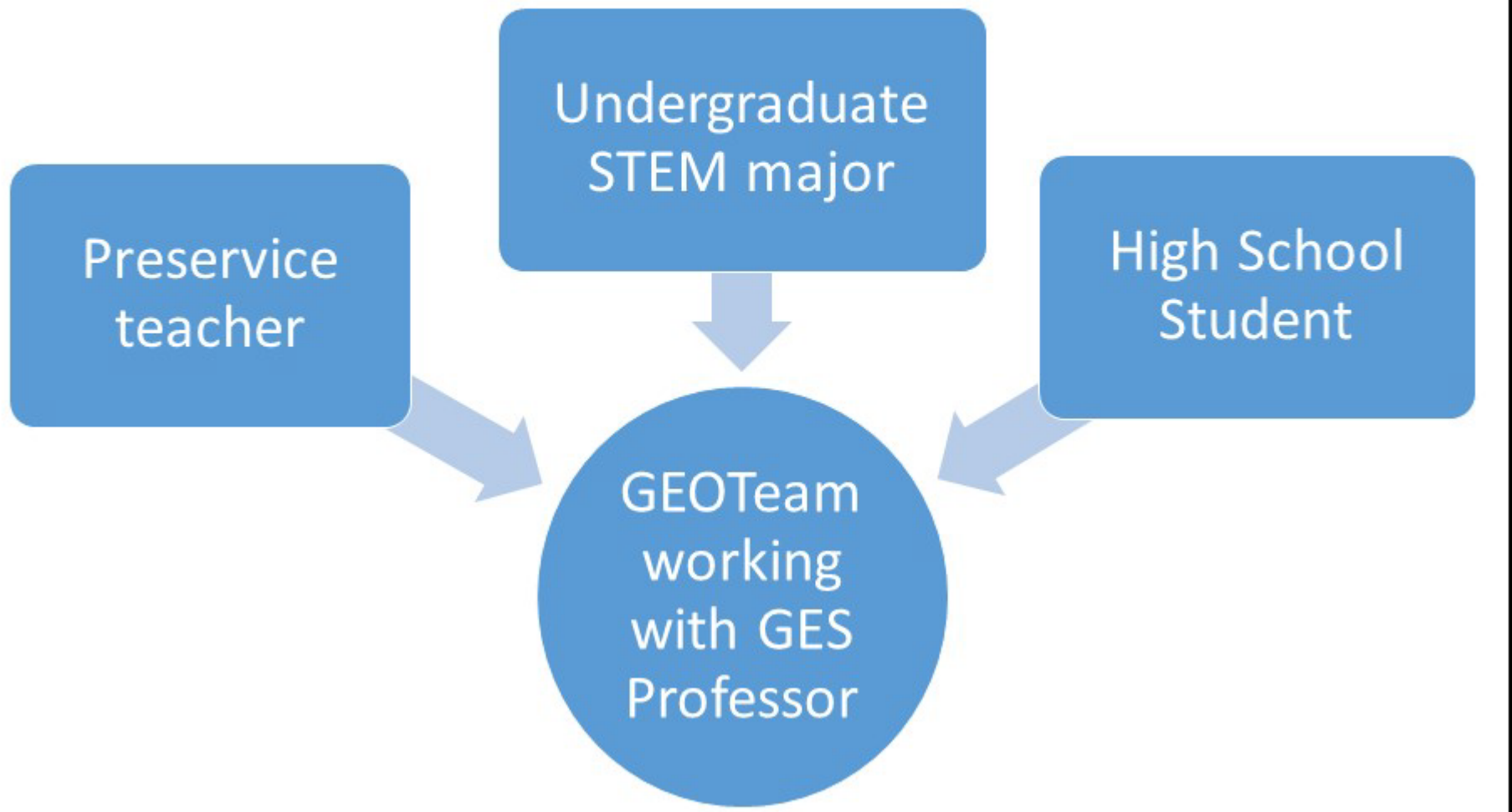


Figure 1. The people that make up each "GEOTeam."

Project Annual Timeline:

April	May	June	Jul/Aug	Sep	Oct - Mar
Advisory Council Meets; plan student recruitment; refine evaluation plan	review applications	assemble GEOTeams	5 week on-campus program	students prepare and present posters	analyze project data and disseminate evaluation outcomes

Collaborative Activities, all GEOTeams together:
A. Weekly Monday workshops:
1. Safety training.
2. Literature review.
3. Research methods and geoscience ethics.
4. Presentation skills.
5. Writing scientific reports.
B. Weekly Wednesday seminars:
1. Overview of geology and environmental science in service of sustainability: overarching research project themes and societal significance.
2. Three local guest speakers from a consulting firm, a non-profit organization, and government; each speak on their work and their individual career paths.
3. Developing students' career journey: college and graduate school admissions.

Individual GEOTeam Activities:
1. Training: lab and field safety; background information; scientific problem design.
2. Development: hypothesis, field and lab methods, assigning tasks to team members, preliminary data collection.
3. Sampling: data collection and lab analysis.
4. Data analysis: review feedback, final sampling, validation.
5. Writing: develop scientific poster.

Figure 2. The approximate annual project timeline.

Results:

In 2021, the evaluation focus was primarily formative. Feedback was provided and used to help optimize project activities. Despite the impact that COVID-19 had on many efforts, the project evaluator found that the project accomplished Goals #1 and #3 (above): "the project team's ability to hold an in-person research experience (masked) during 2021 was impressive" (Hecht, 2022).

Evaluation of the program in the summer of 2022 found that **participants' responses were generally positive**. The focus of evaluation expanded slightly in 2022 to include statements in the surveys about working in teams. Nearly all responses from participants rated statements like "I was comfortable asking my research mentor questions" and "I felt connected with other students in my group" with responses of "Agree" or "Strongly Agree."

The **demographics** of the participants (see **Table 1**) accomplished Goal #4 (above).

Future Plans:

We plan to meet with our **Advisory Board** in January of 2023, to refine our project for the third and final year of GEOTeams in the summer of 2023. We plan to expand the program evaluation of teamwork as an essential component of the GEOTeams program, and assess better the outcomes related to Goals #2 and 5.

We are in the process of writing a chapter for the forthcoming book to be published by the Geological Society of America, titled **GEOPATHS--Broadening the Voices in Geosciences: Pathways to innovative Pedagogy and Professional Development**.

Research by diverse student "GEOTeams" provides effective introductions to geoscience.

Dr. Bennington's GEOTeam investigating Cretaceous fossils in New Jersey, August 2022.



Dr. Marsellos' GEOTeam fixing equipment and retrieving data from Santorini volcano, Greece, July 2022.

Dr. Farmer's GEOTeam visiting Lamont-Doherty Earth Observatory of Columbia University, New York, August 2022.



Dr. Bernhardt's GEOTeam measuring air quality in New York, October 2022.

For more information, please see our website using this QR code:

